

Robin Bowers

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July, 2026

Research interests: Theoretical computer science, algorithmic economics, mechanism design, information aggregation, Pandora’s Box, matching, algorithmic fairness.

EDUCATION

University of Colorado Boulder

Boulder, CO

PhD, Computer Science

2021 – 2026

Thesis title: “Beyond Pandora’s Box: Algorithm and Mechanism Design with Costly Information Acquisition”

Advisors: Bo Waggoner and Rafael Frongillo

University of Colorado Boulder

Boulder, CO

MS, Computer Science

2021 – 2024

Advisors: Bo Waggoner and Rafael Frongillo

Oberlin College

Oberlin, OH

BA, Computer Science and Mathematics

2016 – 2020

University of Edinburgh

Edinburgh, UK

IFSA-Butler Study Abroad

Spring 2019

WORKING PAPERS

Sample Complexity of Peer Prediction.

Abdellah Aznag, Robin Bowers, Rachel Cummings, Jason Hartline, Matthew vonAllmen, Bo Waggoner.

PUBLICATIONS

Authors listed alphabetically unless marked with *.

Combinatorial Markov Search.

(Previously titled Prophet Inequalities for Bandits, Cabinets, and DAGs)

Robin Bowers, Elias Lindgren, Bo Waggoner. *Symposium on Theory of Computing (STOC)*, 2026 (forthcoming).

Polynomial-Time Approximation Schemes via Utility Alignment: Unit-demand Pricing and More.

Robin Bowers, Marius Garbea, Emmanouil Pountourakis, Samuel Taggart, *Symposium on Foundations of Computer Science (FOCS)*, 2025.

Matching with Nested and Bundled Pandora Boxes.

Robin Bowers, Bo Waggoner, *Conference on Web and Internet Economics (WINE)*, 2024.

High-Welfare Matching Markets via Descending Price.

Robin Bowers, Bo Waggoner, *Conference on Web and Internet Economics (WINE)*, 2023.

Loom Pedals: Retooling Jacquard Weaving for Improvisational Design Workflows.*

Shanel Wu, Xavier A Corr, Xi Gao, Sasha De Koninck, Robin Bowers, Laura Devendorf, *Conference on Tangible and Embedded Interaction (TEI)*, 2024.

Machine Learning Based MIMO Equalizer for High Frequency (HF) Communications.*

S. Spillane, K. H. Jung, Bowers, T. Peken, M. H. Marefat and T. Bose, *2020 International Joint Conference on Neural Networks (IJCNN 2020)*.

PROFESSIONAL ACTIVITIES & SERVICE

Participant – EAAMO Inequality Working Group <i>EAAMO Bridges</i>	Spring 2026 – Present
Teaching Assistant – Winter School on Data Economics <i>Moroccan Center for Game Theory, UM6P</i>	January 2025 <i>Rabat, Morocco</i>
• Wrote problem set on value of information and data economics in machine learning • Lead afternoon workshop session	
Co-Organizer – EC Gender Inclusion Workshop <i>Workshop at Conference on Economics and Computation (EC'24, '25)</i>	Summer 2024, 2025 <i>New Haven, CT; Stanford, CA</i>
• Co-organized 2025 workshop on gender inclusion with Natalie Collina, Mirah Shi, and Kunhe Yang • Co-organized 2024 workshop on gender inclusion with Yeganeh Alimohammadi, Natalie Collina, Kate Donahue, Bailey Flanigan, and Maneesha Papireddygar	
Founder/Coordinator – CU Boulder Algorithmic Fairness Reading Group <i>University of Colorado Boulder</i>	Fall 2023 – Fall 2024 <i>Boulder, CO</i>
• Wrote a syllabus of papers for each semester, coordinated discussions of papers	
Participant – SLMath Summer School <i>SLMath</i>	Summer 2023 <i>Berkeley, CA</i>
• Participant in the summer school <u>Mathematics and Computer Science of Market and Mechanism Design</u>	
Coordinator – Algorithmic Economics Reading Group <i>University of Colorado Boulder</i>	2022 – 2023 <i>Boulder, CO</i>
• Planned and scheduled reading group meetings	
CU Boulder Graduate Peer Mentor <i>University of Colorado Boulder</i>	2022 – 2026 <i>Boulder, CO</i>
• Matched with incoming PhD and Master's students through both Graduate School and Computer Science Department mentorship programs • Helped introduce students to research and the CU Boulder course system • Helped international students adjust to Boulder	
CU Boulder Graduate Prospective Student Mentor <i>University of Colorado Boulder</i>	Fall 2022, 2023, 2024 <i>Boulder, CO</i>
• Matched to prospective students applying to graduate schools • Provided feedback on application materials and guidance on CU Boulder's graduate program application process and expectations	
Reviewing EC'26, EAAMO'25, EC'25, FAccT'25, EAAMO'24	

AWARDS

Department Research Expo – Published Work Award <i>University of Colorado Boulder, Computer Science Department</i>	2026 <i>Boulder, CO</i>
Outstanding Service Award <i>University of Colorado Boulder, Computer Science Department</i>	2024 <i>Boulder, CO</i>
Outstanding Teaching Assistant Award <i>University of Colorado Boulder, Computer Science Department</i>	2022 <i>Boulder, CO</i>
Department Research Expo – Work In Progress Award <i>University of Colorado Boulder, Computer Science Department</i>	2022 <i>Boulder, CO</i>
R.J. Thomas Computer Science Teaching Assistant Award <i>Oberlin College</i>	May 2020 <i>Oberlin, OH</i>
John F. Oberlin Scholarship <i>Oberlin College</i>	2016 – 2020 <i>Oberlin, OH</i>

TEACHING

Guest Lecture – Discrete Math

CU Boulder

Professor Bo Waggoner

Fall 2024

Boulder, CO

Two Guest Lectures – Graduate Algorithms

CU Boulder

Professor Sriram Sankaranarayanan

Fall 2024

Boulder, CO

Teaching Assistant – Graduate Algorithms

University of Colorado Boulder

- Contributed to writing homework solutions and grading, held weekly office hours.

Fall 2024

Boulder, CO

Teaching Assistant – Principles of Programming Languages

University of Colorado Boulder

- Led weekly practice sessions on functional programming and programming language construction

Fall 2023

Boulder, CO

Teaching Assistant – Algorithms

University of Colorado Boulder

- Led weekly practice sessions on course topics
- Wrote weekly problem sets and solutions presented in all TA sections
- Received a departmental teaching assistant award for my contributions

Spring 2022

Boulder, CO

Teaching Assistant – Algorithms

Oberlin College

- Wrote worksheets for and led weekly practice sessions on course topics
- Transferred problem session material online for distance learning
- Received a departmental teaching assistant award for my contributions

2019 – 2020

Oberlin, OH

Grader – Discrete Math

Oberlin College

- Graded homework and quizzes weekly for class size of 30-80 students for in-person and distance learning classes
- Provided written feedback to students on mathematical reasoning and clarity

Fall 2018, Spring 2020

Oberlin, OH

INVITED TALKS

Polynomial-Time Approximation Schemes via Utility Alignment: Unit-demand Pricing and More.

- UT Austin Theory Seminar (January 2026)

Prophet Inequalities for Bandits, Cabinets, and DAGs.

- Boston College Theory Seminar (July 2025)
- UT Austin Theory Seminar (June 2025)

POSTERS & TALKS

Polynomial-Time Approximation Schemes via Utility Alignment: Unit-demand Pricing and More.

Robin Bowers, Marius Garbea, Emmanouil Pountourakis, Samuel Taggart, Talk presented at FOCS 2025.

Matching with Nested and Bundled Pandora Boxes

Robin Bowers, Bo Waggoner, Poster presented at ACM Conference on Economics and Computation (EC), 2024; Talk presented at WINE 2024.

High-Welfare Matching Markets via Descending Price

Robin Bowers, Bo Waggoner, Poster presented at Marketplace Innovation Workshop, 2023; Simons Institute Workshop on Societal Considerations and Applications, 2022; ACM Conference on Economics and Computation, 2022. Talk presented at WINE 2023.

OTHER EXPERIENCE

Undergraduate Research Assistant

2019 – 2020

Oberlin College

Oberlin, OH

- Conducted research in economic game theory
- Proved setting-specific performance lower-bounds for several auction algorithms
- Obtained complexity results on computing lower bounds for performance of auctions in specific settings

Research Internship – NSF REU

Summer 2019

University of Arizona Department of Electrical and Computer Engineering

Tucson, AZ

- Conducted study on “Machine Learning Based MIMO Equalizer for High Frequency (HF) Communications”
- Created mathematical models of long-range radio systems in MATLAB
- Implemented machine-learning algorithms to dynamically adjust system behavior for best performance
- Communicated professionally with other students to coordinate project and goals

Software Internship

Summer 2018

Blendid

Sunnyvale, CA

- Worked with Universal Robots robotic arm designing and optimizing movement paths in automated food kiosk
- Handled emergency kiosk malfunctions and daily prototype operations